Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

Submission Date: / / 25

2nd Year 1st Semester (Hons)

Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No:

Experiment Title:

Submitted To:

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Experiment No: 1

Experiment Name: Verification of basic logic gates

Objectives:

- To verify the truth tables of basic logic gates (AND, OR, NAND, NOR, NOT)
- To understand the functionality of each gate
- To implement logic gates using ICs and test them using a bread board.

Apparatus:

Breadboard, connecting wires, digital trainer board.

IC 7408 (AND Gate)
IC 7432 (OR Gate)
IC 7400 (NAND Gate)
IC 7402 (NOR Gate)
IC 7404 (NOT Gate)

Verification of truth tables:

- AND Gate:

a	b	a.b
0	0	0
0	1	0
1	0	0
1	1	1

- OR Gate:

a	b	a+b
0	0	0
0	1	1
1	0	1
1	1	1

P.T.O

- NAND Gate:

a	b	$\overline{a.b}$
0	0	1
0	1	1
1	0	1
1	1	0

- NOR Gate

a	b	$\overline{a+b}$
0	0	1
0	1	0
1	0	0
1	1	0

- NOT Gate

a	$\overline{a}$
0	1
1	0

Diagram:

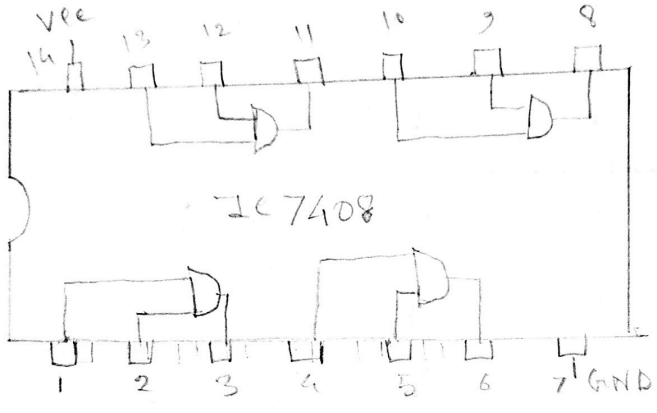


Fig: IC 7408 Pin diagram

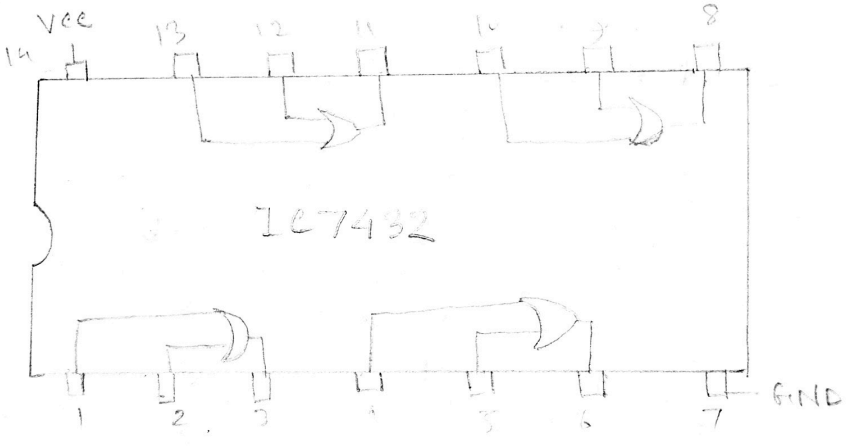


Fig: IC 7432 Pin diagram

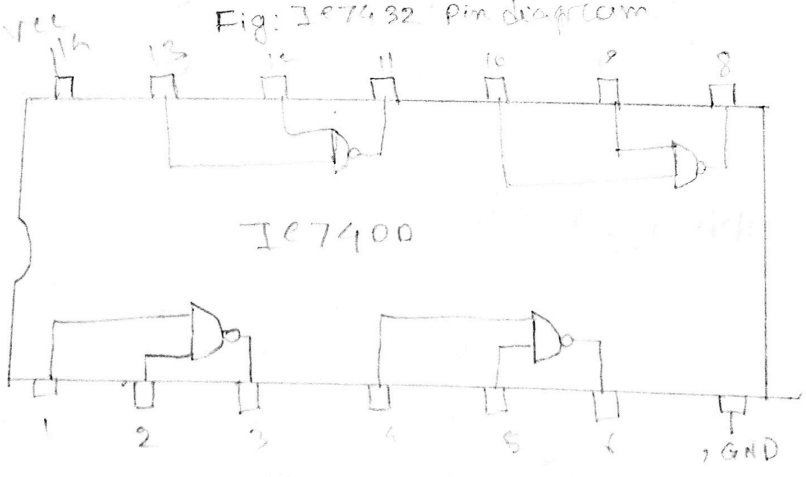


Fig: IC 7400 Pin diagram

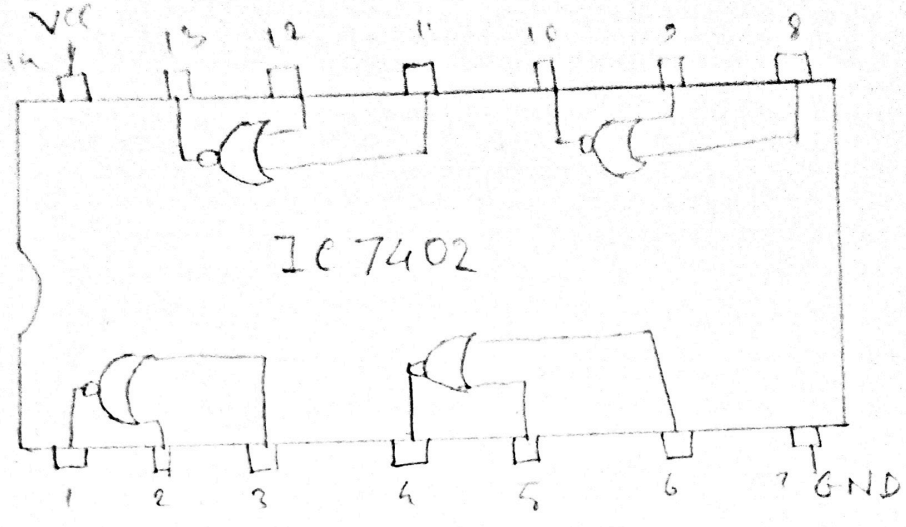


Fig: IC 7402 Pin diagram

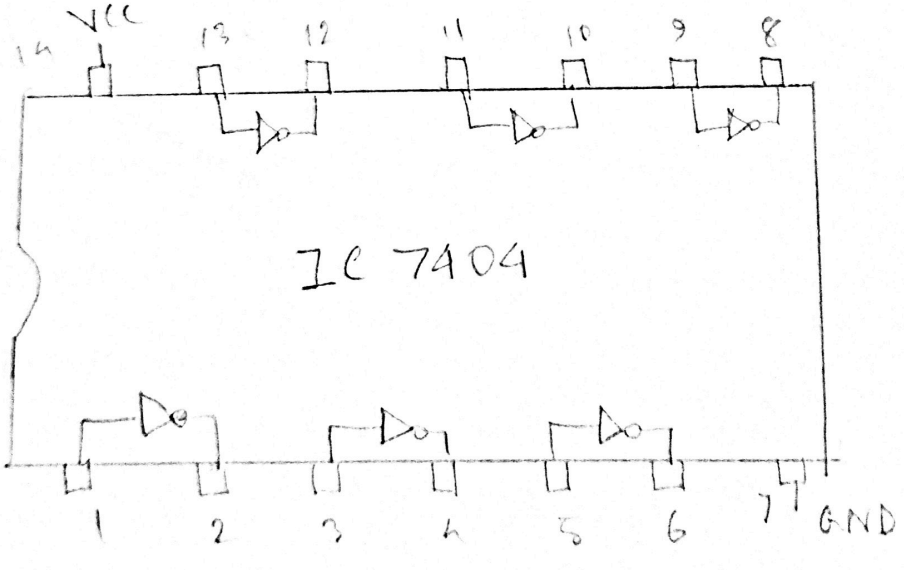


Fig: IC 7404 Pin diagram